

BioCoLoop: STATISTICAL SUPPLEMENT

Anonymous authors

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This supplement accompanies *BioCoLoop: Collaborative Agentic Research for Biological Model Improvement*. It reports all per-seed comparisons from the two complete uncertainty tables, without selecting results by direction or significance. Tables S.1 and S.2 retain their original numerical records. Mean scores, the full six-arm comparison and the decomposition of effects remain in Appendix A of the manuscript.

Intervals are paired cluster-bootstrap 95% intervals from 1,000 fixed-seed resamples, conditional on the selected trained predictors. Drug identity defines clusters for random and unseen-drug interaction tests, protein sequence for unseen-protein tests, and compound or intervention identity for proteomic or cell tests. Each contrast uses the same sampled clusters for both methods. Seeds 42–44 use fixed data partitions. These intervals measure test-sample uncertainty, separately from training-seed variation.

Table S.1: Paired cluster-bootstrap 95% intervals for each seed and main contrast, in percentage points. Intervals condition on the selected trained models; DTI random-endpoint rows resample drug clusters.

Task	Contrast	Seed	Difference	95% interval
DTI	Loop – fixed (10 labs)	42	+0.33	[-0.06, +0.69]
DTI	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
DTI	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	42	+12.51	[+11.35, +13.71]
DTI	10 – 1 labs (both direct)	42	+5.78	[+5.02, +6.57]
DTI	10 – 1 labs (both loop)	42	+5.78	[+5.02, +6.57]
DTI	Loop – fixed (10 labs)	43	+0.54	[-0.08, +1.11]
DTI	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
DTI	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	43	+7.80	[+6.82, +8.82]
DTI	10 – 1 labs (both direct)	43	+1.65	[+0.90, +2.38]
DTI	10 – 1 labs (both loop)	43	+1.65	[+0.90, +2.38]
DTI	Loop – fixed (10 labs)	44	+5.01	[+4.25, +5.71]
DTI	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
DTI	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
DTI	10 – 1 labs (both fixed)	44	+6.60	[+5.61, +7.74]
DTI	10 – 1 labs (both direct)	44	+4.17	[+3.45, +4.93]
DTI	10 – 1 labs (both loop)	44	+4.17	[+3.45, +4.93]
Proteomics	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	42	+13.16	[-0.39, +26.02]
Proteomics	10 – 1 labs (both direct)	42	+16.91	[+3.37, +31.02]
Proteomics	10 – 1 labs (both loop)	42	+16.91	[+3.37, +31.02]
Proteomics	Loop – fixed (10 labs)	43	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	43	+15.98	[+1.29, +29.00]
Proteomics	10 – 1 labs (both direct)	43	+17.03	[+3.34, +31.44]
Proteomics	10 – 1 labs (both loop)	43	+17.03	[+3.34, +31.44]

Task	Contrast	Seed	Difference	95% interval
Proteomics	Loop – fixed (10 labs)	44	-0.24	[-3.40, +2.33]
Proteomics	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Proteomics	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Proteomics	10 – 1 labs (both fixed)	44	+6.93	[-3.43, +15.60]
Proteomics	10 – 1 labs (both direct)	44	+16.63	[+3.02, +31.08]
Proteomics	10 – 1 labs (both loop)	44	+16.63	[+3.02, +31.08]
VCC	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
VCC	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	42	+1.70	[-5.36, +8.58]
VCC	10 – 1 labs (both direct)	42	+9.90	[-2.88, +23.04]
VCC	10 – 1 labs (both loop)	42	+9.90	[-2.88, +23.04]
VCC	Loop – fixed (10 labs)	43	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	43	+4.83	[-5.50, +16.49]
VCC	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	43	+1.56	[-5.76, +8.58]
VCC	10 – 1 labs (both direct)	43	+15.08	[+2.94, +27.78]
VCC	10 – 1 labs (both loop)	43	+10.24	[-2.74, +23.49]
VCC	Loop – fixed (10 labs)	44	+0.00	[+0.00, +0.00]
VCC	Loop – direct (1 lab)	44	-7.44	[-17.23, +1.33]
VCC	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
VCC	10 – 1 labs (both fixed)	44	+1.56	[-5.76, +8.58]
VCC	10 – 1 labs (both direct)	44	+6.38	[-4.38, +19.13]
VCC	10 – 1 labs (both loop)	44	+13.82	[+2.54, +26.34]
Norman	Loop – fixed (10 labs)	42	+7.26	[-9.41, +25.41]
Norman	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	42	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	42	+8.59	[-8.30, +26.00]
Norman	10 – 1 labs (both loop)	42	+8.59	[-8.30, +26.00]
Norman	Loop – fixed (10 labs)	43	+6.37	[-11.12, +24.81]
Norman	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	43	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	43	+7.70	[-9.33, +25.85]
Norman	10 – 1 labs (both loop)	43	+7.70	[-9.33, +25.85]
Norman	Loop – fixed (10 labs)	44	+6.67	[-10.37, +24.81]
Norman	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Norman	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Norman	10 – 1 labs (both fixed)	44	-10.07	[-21.19, +0.00]
Norman	10 – 1 labs (both direct)	44	+8.00	[-8.52, +25.93]
Norman	10 – 1 labs (both loop)	44	+8.00	[-8.52, +25.93]
Tahoe	Loop – fixed (10 labs)	42	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (1 lab)	42	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	42	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	42	+4.48	[-5.97, +14.93]
Tahoe	10 – 1 labs (both direct)	42	+10.45	[-1.49, +22.39]
Tahoe	10 – 1 labs (both loop)	42	+10.45	[-1.49, +22.39]
Tahoe	Loop – fixed (10 labs)	43	-2.99	[-7.46, +0.00]
Tahoe	Loop – direct (1 lab)	43	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	43	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	43	+7.46	[-4.48, +17.95]
Tahoe	10 – 1 labs (both direct)	43	+13.43	[+1.49, +23.88]

Task	Contrast	Seed	Difference	95% interval
Tahoe	10 – 1 labs (both loop)	43	+13.43	[+1.49, +23.88]
Tahoe	Loop – fixed (10 labs)	44	-7.46	[-19.40, +4.48]
Tahoe	Loop – direct (1 lab)	44	+0.00	[+0.00, +0.00]
Tahoe	Loop – direct (10 labs)	44	+0.00	[+0.00, +0.00]
Tahoe	10 – 1 labs (both fixed)	44	+7.46	[-4.48, +17.95]
Tahoe	10 – 1 labs (both direct)	44	-2.99	[-17.91, +11.94]
Tahoe	10 – 1 labs (both loop)	44	-2.99	[-17.91, +11.94]

Table S.2: Post-hoc TAPB paired AUROC/AP differences and 95% intervals from 1,000 fixed-seed cluster-bootstrap draws. Random and unseen-drug endpoints use drug clusters; unseen-protein uses protein clusters. All three endpoints, six arms and prespecified contrasts are shown.

Endpoint / seed	Contrast	AUROC Δ	95% interval	AP Δ	95% interval
Random / 42	Loop — fixed (10 labs)	+0.33	[-0.06, +0.69]	+1.40	[+0.76, +2.05]
Random / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 42	10 — 1 labs (fixed)	+12.51	[+11.35, +13.71]	+10.56	[+9.10, +12.11]
Random / 42	10 — 1 labs (direct)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Random / 42	10 — 1 labs (loop)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Random / 42	Our harness — fixed (1 lab)	+12.84	[+11.82, +14.01]	+11.96	[+10.56, +13.45]
Random / 42	Our harness — direct (1 lab)	+5.78	[+5.02, +6.57]	+8.29	[+7.01, +9.56]
Unseen drug / 42	Loop — fixed (10 labs)	+0.10	[-0.47, +0.67]	+1.06	[+0.42, +1.66]
Unseen drug / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 42	10 — 1 labs (fixed)	+12.30	[+10.77, +13.98]	+7.17	[+5.82, +8.72]
Unseen drug / 42	10 — 1 labs (direct)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen drug / 42	10 — 1 labs (loop)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen drug / 42	Our harness — fixed (1 lab)	+12.40	[+10.94, +13.85]	+8.22	[+6.97, +9.66]
Unseen drug / 42	Our harness — direct (1 lab)	+6.38	[+5.34, +7.53]	+7.55	[+6.30, +8.92]
Unseen protein / 42	Loop — fixed (10 labs)	+0.03	[-5.53, +4.28]	+4.12	[-4.10, +8.22]
Unseen protein / 42	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 42	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 42	10 — 1 labs (fixed)	+20.02	[+0.16, +63.83]	+8.14	[-8.27, +51.77]
Unseen protein / 42	10 — 1 labs (direct)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Unseen protein / 42	10 — 1 labs (loop)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Unseen protein / 42	Our harness — fixed (1 lab)	+20.05	[+1.84, +61.27]	+12.25	[-3.01, +53.19]
Unseen protein / 42	Our harness — direct (1 lab)	+8.86	[+2.48, +13.39]	+13.26	[+3.15, +28.50]
Random / 43	Loop — fixed (10 labs)	+0.54	[-0.08, +1.11]	+0.44	[-0.54, +1.44]
Random / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 43	10 — 1 labs (fixed)	+7.80	[+6.82, +8.82]	+7.68	[+6.32, +9.13]
Random / 43	10 — 1 labs (direct)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Random / 43	10 — 1 labs (loop)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Random / 43	Our harness — fixed (1 lab)	+8.35	[+7.29, +9.37]	+8.12	[+6.53, +9.76]
Random / 43	Our harness — direct (1 lab)	+1.65	[+0.90, +2.38]	+1.17	[-0.11, +2.50]
Unseen drug / 43	Loop — fixed (10 labs)	-0.51	[-1.44, +0.29]	-1.10	[-2.16, -0.18]
Unseen drug / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 43	10 — 1 labs (fixed)	+6.68	[+5.33, +8.25]	+4.66	[+3.33, +6.36]
Unseen drug / 43	10 — 1 labs (direct)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen drug / 43	10 — 1 labs (loop)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen drug / 43	Our harness — fixed (1 lab)	+6.17	[+4.85, +7.48]	+3.57	[+2.10, +5.09]
Unseen drug / 43	Our harness — direct (1 lab)	-0.15	[-0.96, +0.67]	-1.43	[-2.43, -0.42]
Unseen protein / 43	Loop — fixed (10 labs)	-0.96	[-7.24, +3.07]	-1.83	[-21.55, +7.54]
Unseen protein / 43	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 43	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 43	10 — 1 labs (fixed)	+4.79	[-6.61, +28.12]	+0.41	[-16.48, +31.69]
Unseen protein / 43	10 — 1 labs (direct)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Unseen protein / 43	10 — 1 labs (loop)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Unseen protein / 43	Our harness — fixed (1 lab)	+3.83	[-5.83, +22.32]	-1.42	[-14.49, +16.88]
Unseen protein / 43	Our harness — direct (1 lab)	-1.19	[-6.22, +6.09]	-6.76	[-13.95, +6.71]
Random / 44	Loop — fixed (10 labs)	+5.01	[+4.25, +5.71]	+8.28	[+7.07, +9.56]
Random / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Random / 44	10 — 1 labs (fixed)	+6.60	[+5.61, +7.74]	+5.12	[+3.71, +6.80]
Random / 44	10 — 1 labs (direct)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Random / 44	10 — 1 labs (loop)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Random / 44	Our harness — fixed (1 lab)	+11.60	[+10.60, +12.65]	+13.40	[+11.88, +15.08]
Random / 44	Our harness — direct (1 lab)	+4.17	[+3.45, +4.93]	+5.44	[+4.23, +6.64]
Unseen drug / 44	Loop — fixed (10 labs)	+5.18	[+4.09, +6.19]	+6.57	[+5.28, +7.85]
Unseen drug / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen drug / 44	10 — 1 labs (fixed)	+5.32	[+3.81, +6.97]	+2.04	[+0.38, +3.90]
Unseen drug / 44	10 — 1 labs (direct)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]
Unseen drug / 44	10 — 1 labs (loop)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]
Unseen drug / 44	Our harness — fixed (1 lab)	+10.50	[+9.20, +11.87]	+8.61	[+7.10, +10.29]
Unseen drug / 44	Our harness — direct (1 lab)	+3.54	[+2.69, +4.48]	+3.57	[+2.55, +4.72]
Unseen protein / 44	Loop — fixed (10 labs)	+6.42	[-2.79, +17.37]	+16.15	[-5.79, +26.26]
Unseen protein / 44	Loop — direct (1 lab)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 44	Loop — direct (10 labs)	+0.00	[+0.00, +0.00]	+0.00	[+0.00, +0.00]
Unseen protein / 44	10 — 1 labs (fixed)	+3.39	[-16.02, +33.81]	-8.03	[-26.82, +16.99]
Unseen protein / 44	10 — 1 labs (direct)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]
Unseen protein / 44	10 — 1 labs (loop)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]

Endpoint / seed	Contrast	AUROC Δ	95% interval	AP Δ	95% interval
Unseen protein / 44	Our harness — fixed (1 lab)	+9.81	[-2.07, +34.89]	+8.12	[-12.06, +22.71]
Unseen protein / 44	Our harness — direct (1 lab)	+5.45	[+0.61, +8.74]	+7.10	[-8.65, +12.63]